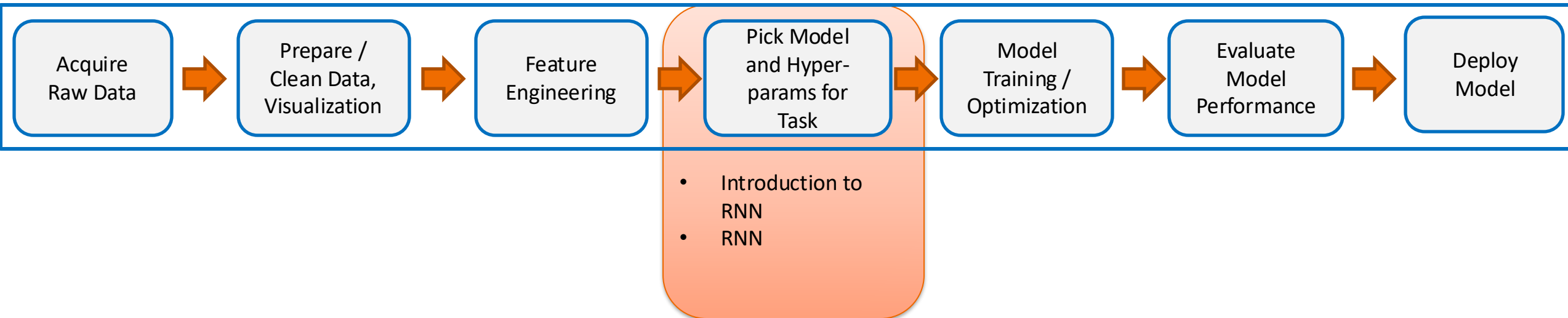
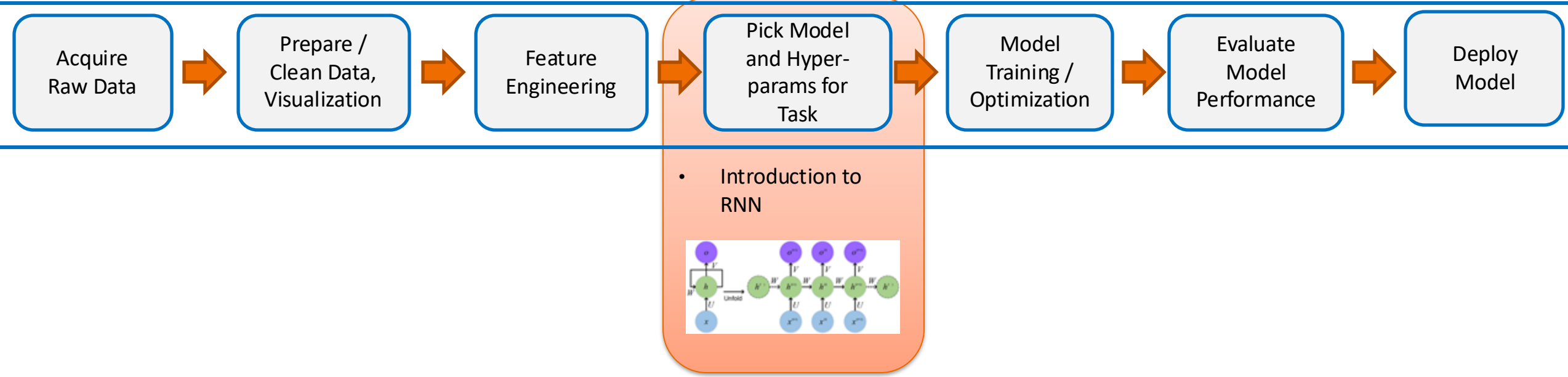


Focus for this lecture



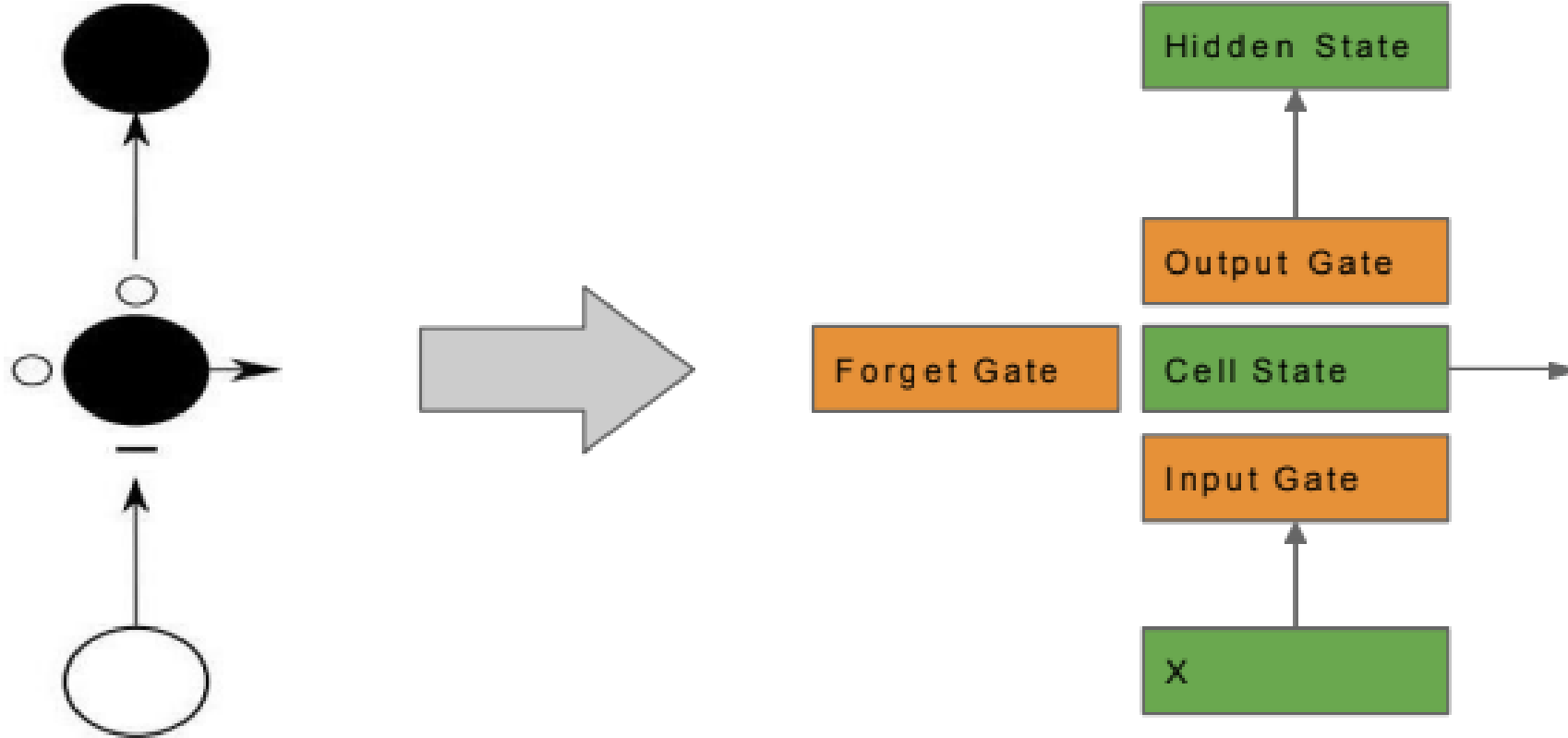


Introduction to RNNs

Why RNNs ?

- Intelligent systems (Networks) need memory.
- Many inputs are sequential in nature.
- Concepts have long term dependencies.
 - Not just one or two steps backwards.
 - Eg. What controls tomato price of tomorrow?
- Popular networks (Eg. CNNs) do not have cycles.

Cell with Memory



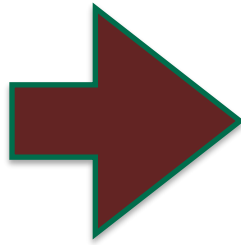
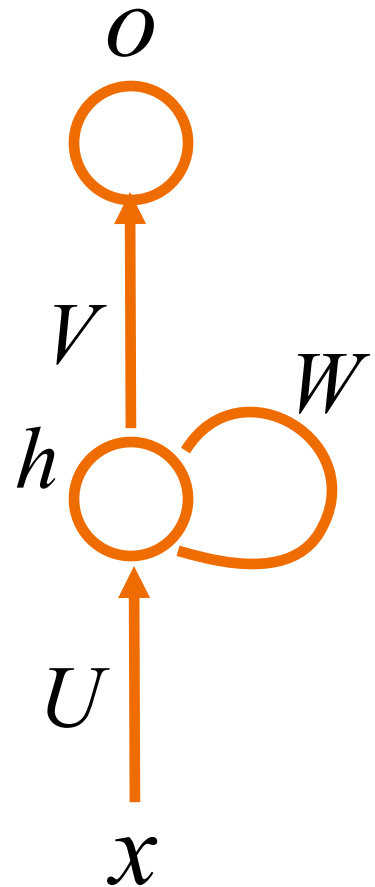
Generating poetry with RNNs

Sonnet 116 – Let me not ...

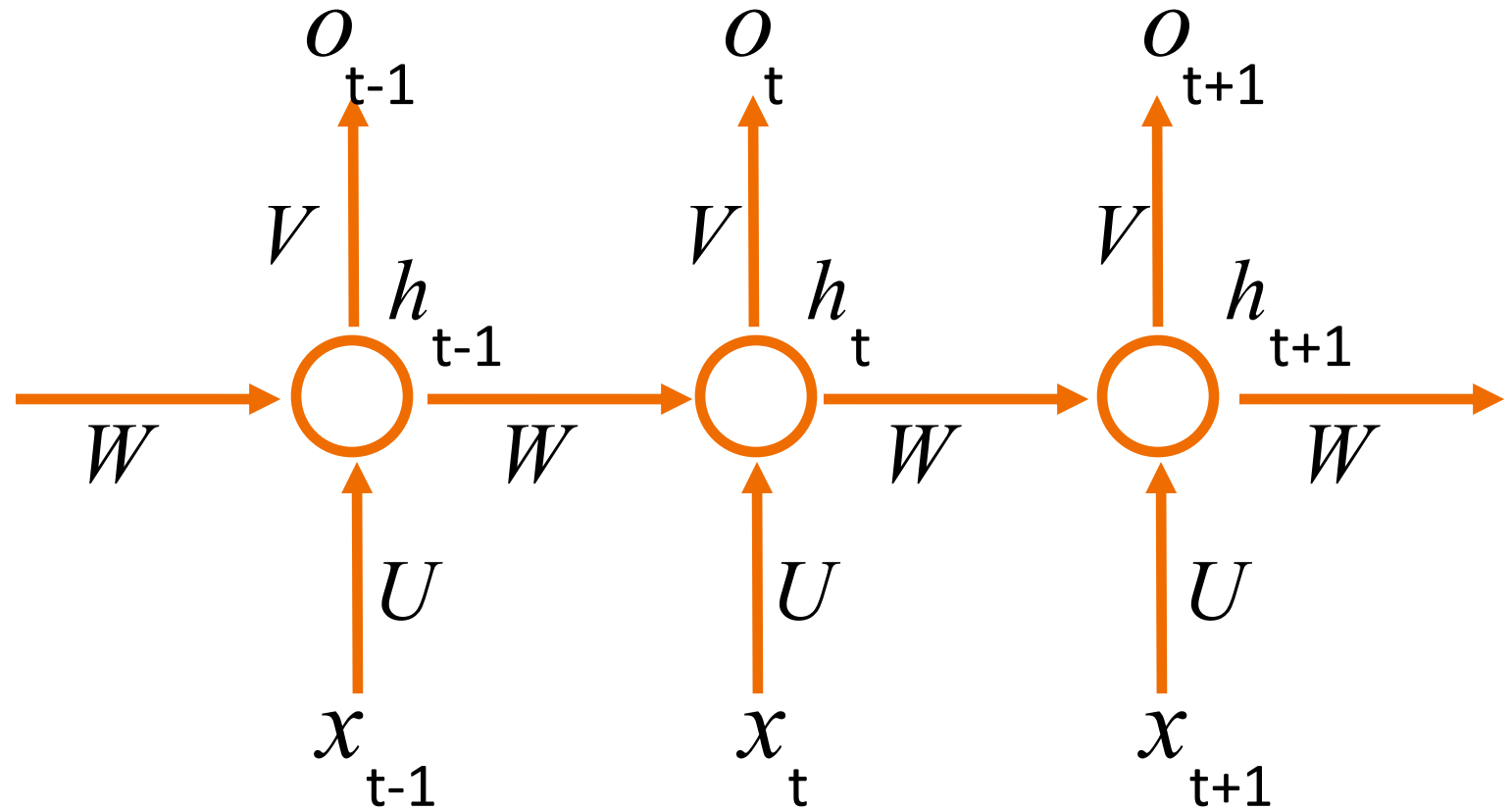
by William Shakespeare

Let me not to the marriage of true minds
 Admit impediments. Love is not love
Which alters when it alteration finds,
 Or bends with the remover to remove:
O no! it is an ever-fixed mark
 That looks on tempests and is never shaken;
It is the star to every wandering bark,
 Whose worth's unknown, although his height be taken.
Love's not Time's fool, though rosy lips and cheeks
 Within his bending sickle's compass come:
Love alters not with his brief hours and weeks,
 But bears it out even to the edge of doom.
If this be error and upon me proved,
 I never writ, nor no man ever loved.

RNN basic architecture



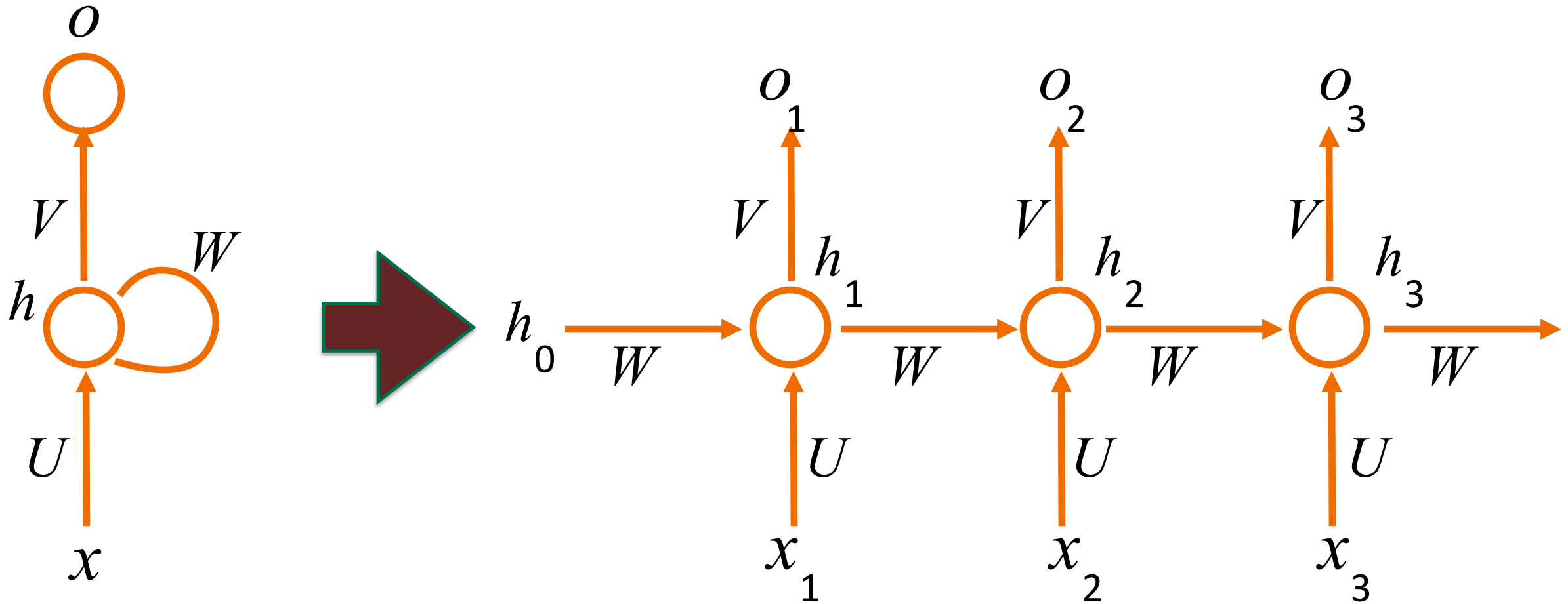
Training through back propagation



$$h_t = f(Ux_t + Wh_{t-1})$$

$$o_t = \text{softmax}(Vh_t)$$

RNN basic architecture



$$h_t = f(Ux_t + Wh_{t-1})$$

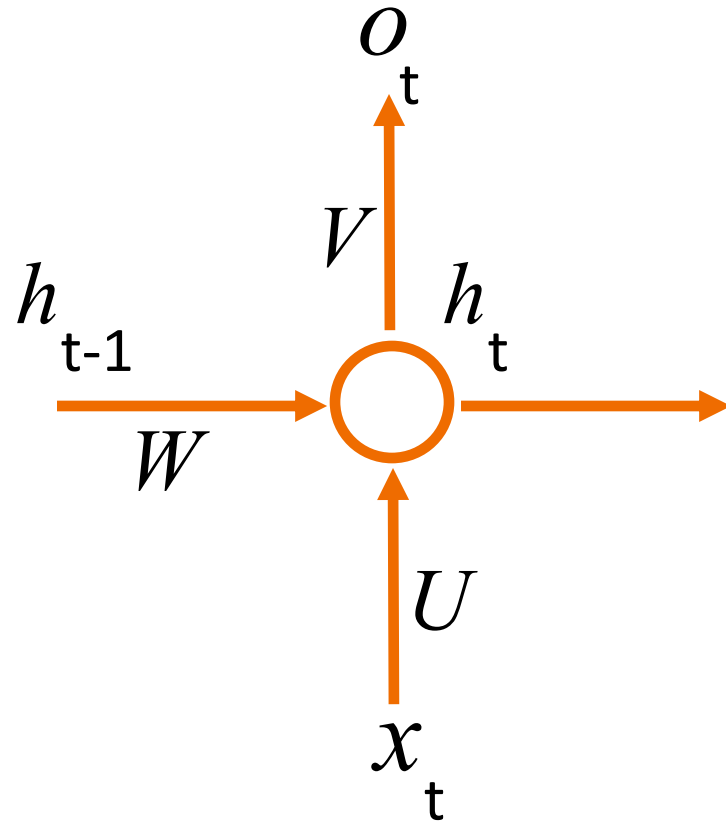
$$o_t = \text{softmax}(Vh_t)$$

RNN basic architecture

- x_t - input at time step t
- h_t - hidden state at time step t (memory of the network)
- o_t - output at time step t
- U, V, W are parameters (shared across all layers)

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RNN basic architecture



$$h_t = f(Ux_t + Wh_{t-1})$$

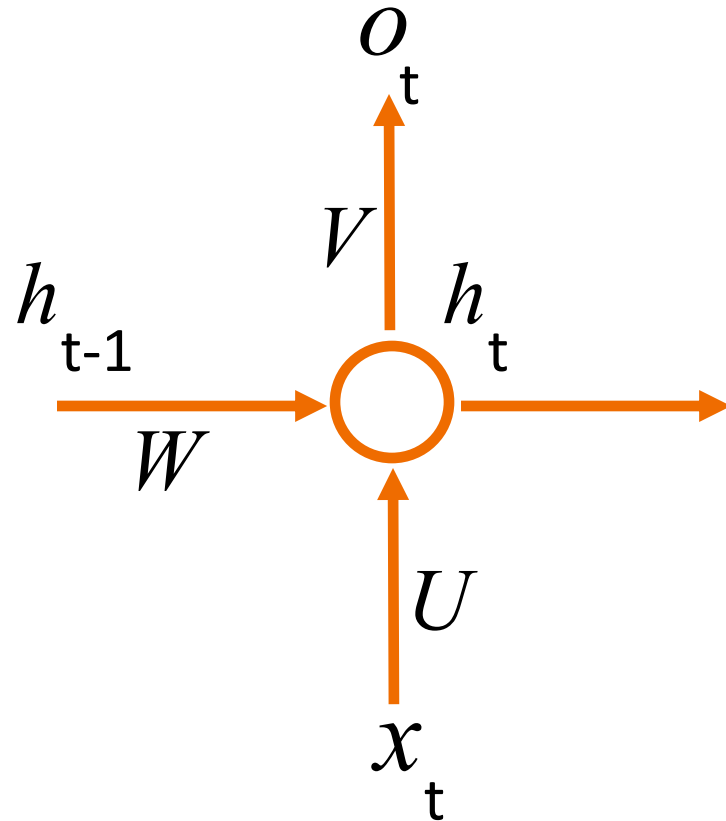
$$U = \begin{pmatrix} 0.5 & 0.4 \\ 0.4 & 0.6 \end{pmatrix}$$

$$W = \begin{pmatrix} 0.4 & 0.5 \\ 0.3 & 0.5 \end{pmatrix} \quad x_t = \begin{pmatrix} 0.4 \\ 0.2 \end{pmatrix} \quad h_{t-1} = \begin{pmatrix} 0.3 \\ 0.8 \end{pmatrix}$$

$$h_t = \tanh \left(\begin{matrix} U & x_t \\ \begin{pmatrix} 0.5 & 0.4 \\ 0.4 & 0.6 \end{pmatrix} & \begin{pmatrix} 0.4 \\ 0.2 \end{pmatrix} \end{matrix} + \begin{matrix} W & h_{t-1} \\ \begin{pmatrix} 0.4 & 0.5 \\ 0.3 & 0.5 \end{pmatrix} & \begin{pmatrix} 0.3 \\ 0.8 \end{pmatrix} \end{matrix} \right)$$

$$h_t = \tanh \left(\begin{pmatrix} .28 \\ .28 \end{pmatrix} + \begin{pmatrix} .52 \\ .29 \end{pmatrix} \right) = \tanh \begin{pmatrix} .88 \\ .77 \end{pmatrix} = \begin{pmatrix} .66 \\ .64 \end{pmatrix}$$

RNN basic architecture



$$U = \begin{pmatrix} 0.5 & 0.4 \\ 0.4 & 0.6 \end{pmatrix}$$

$$W = \begin{pmatrix} 0.4 & 0.5 \\ 0.3 & 0.5 \end{pmatrix} \quad x_t = \begin{pmatrix} 0.4 \\ 0.2 \end{pmatrix} \quad h_{t-1} = \begin{pmatrix} 0.3 \\ 0.8 \end{pmatrix}$$

$$h_t = \begin{pmatrix} .66 \\ .64 \end{pmatrix} \quad V = \begin{pmatrix} 0.4 & 0.7 \\ 0.3 & 0.1 \end{pmatrix}$$

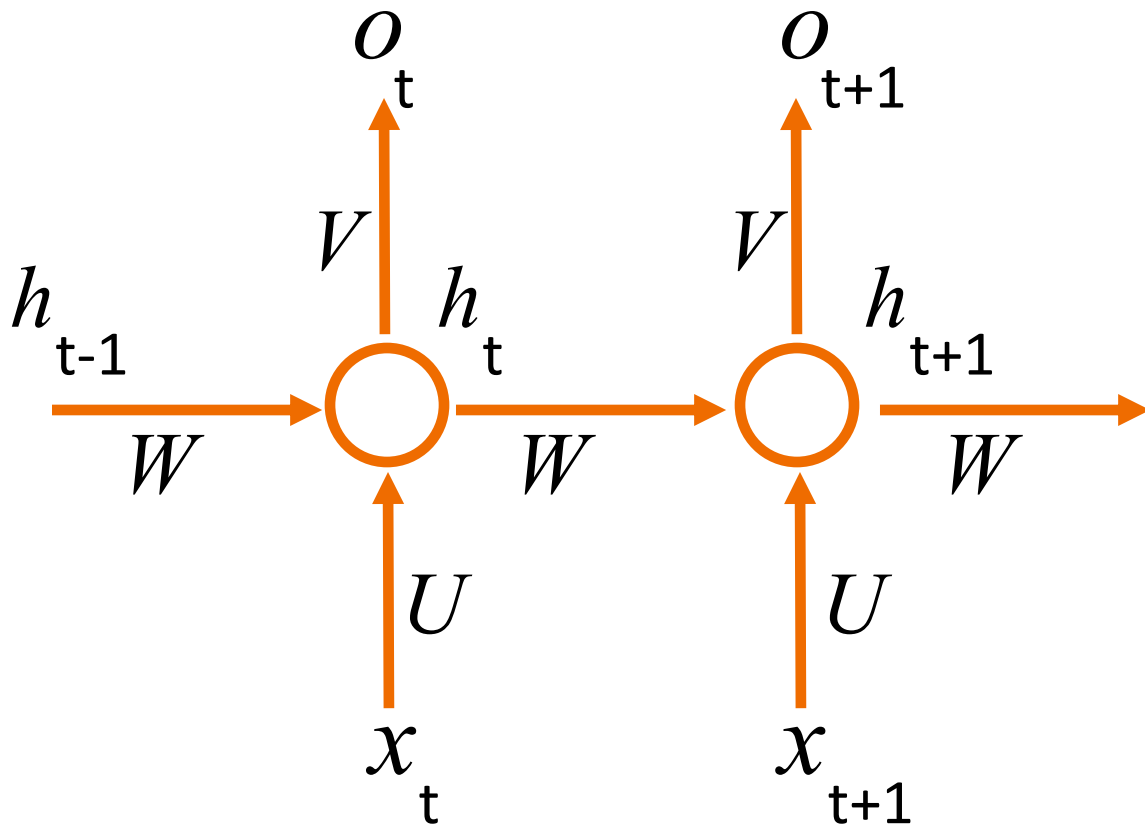
$$o_t = \text{softmax} \left(\begin{pmatrix} 0.4 & 0.7 \\ 0.3 & 0.1 \end{pmatrix} \begin{pmatrix} .66 \\ .64 \end{pmatrix} \right) = \begin{pmatrix} .61 \\ .39 \end{pmatrix}$$

$V \qquad h_t$

$$h_t = f(Ux_t + Wh_{t-1})$$

$$o_t = \text{softmax}(Vh_t)$$

RNN basic architecture



$$h_{t+1} = f(Ux_{t+1} + Wh_t)$$

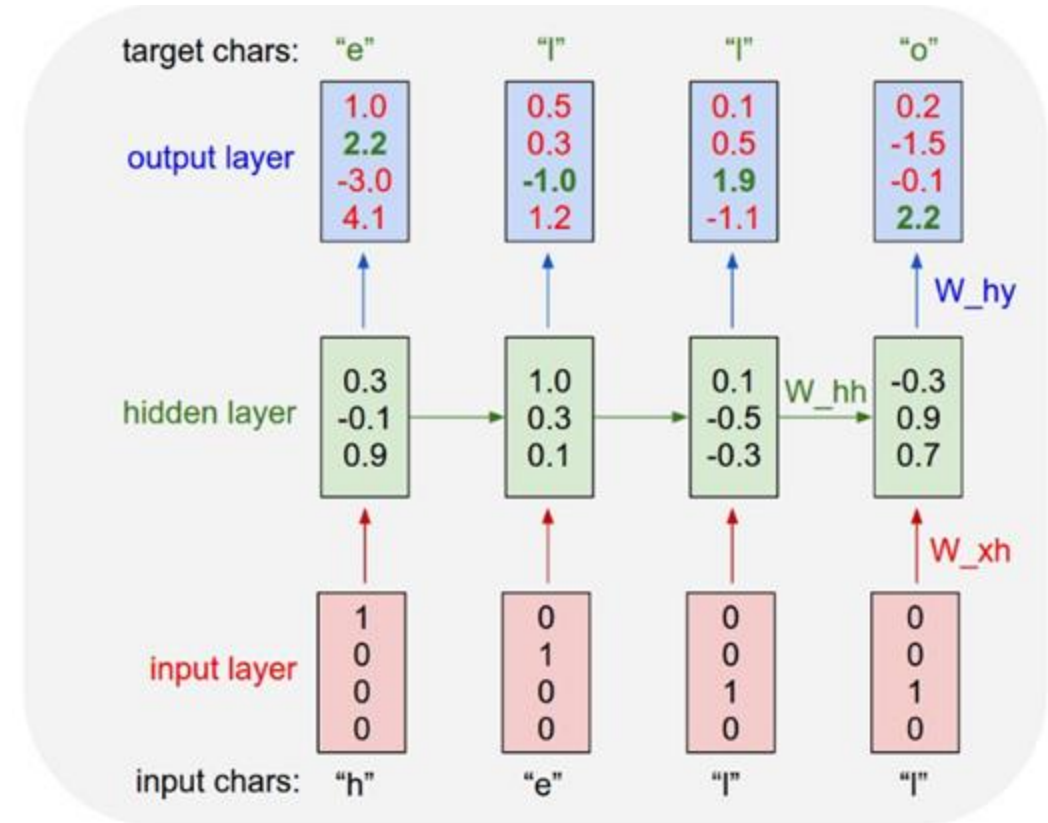
$$U = \begin{pmatrix} 0.5 & 0.4 \\ 0.4 & 0.6 \end{pmatrix}$$
$$W = \begin{pmatrix} 0.4 & 0.5 \\ 0.3 & 0.5 \end{pmatrix} \quad x_{t+1} = \begin{pmatrix} 0.5 \\ 0.4 \end{pmatrix} \quad h_t = \begin{pmatrix} .66 \\ .64 \end{pmatrix}$$
$$h_{t+1} = \tanh \left(\begin{matrix} U & x_{t+1} \\ \begin{pmatrix} 0.5 & 0.4 \\ 0.4 & 0.6 \end{pmatrix} & \begin{pmatrix} 0.5 \\ 0.4 \end{pmatrix} \end{matrix} + \begin{matrix} W & h_t \\ \begin{pmatrix} 0.4 & 0.5 \\ 0.3 & 0.5 \end{pmatrix} & \begin{pmatrix} .66 \\ .64 \end{pmatrix} \end{matrix} \right)$$

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Character Level Language Modelling

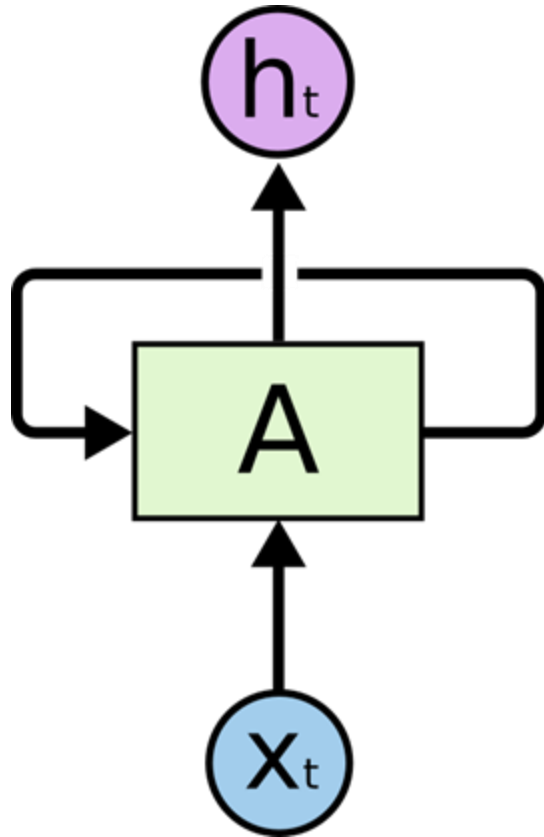
Task:

- Predicting the next character given the current character



It may be today

- RNNs



```
*  
* Increment the size file of the new incorrect UI_FILTER  
* of the size generatively.  
*/  
static int indicate_policy(void)  
  
int error;  
if (fd == MARN_EPT) {  
    /*  
     * The kernel blank will coeld it to userspace.  
     */  
    if (ss->segment < mem_total)  
        unblock_graph_and_set_blocked();  
    else  
        ret = 1;  
    goto bail;  
}  
segaddr = in_SB(in.addr);  
selector = seg / 16;  
setup_works = true;  
for (i = 0; i < blocks; i++) {  
    seq = buf[i++];  
    bpf = bd->bd.next + i * search;  
    if (fd) {  
        current = blocked;
```

And Shakespeare

Second Senator:

They are away this miseries, produced upon my soul,
Breaking and strongly should be buried, when I perish
The earth and thoughts of many states.

DUKE VINCENTIO:

Well, your wit is in the care of side and that.

Second Lord:

They would be ruled after this chamber, and
my fair nues begun out of the fact, to be conveyed,
Whose noble souls I'll have the heart of the wars.

Clown:

Come, sir, I will make did behold your worship.

and Algebraic Geometry!!

Lemma 0.1. Assume (3) and (3) by the construction in the description.

Suppose $X = \lim |X|$ (by the formal open covering X and a single map $\text{Proj}_X(\mathcal{A}) = \text{Spec}(B)$ over U compatible with the complex

$$\text{Set}(\mathcal{A}) = \Gamma(X, \mathcal{O}_X, \mathcal{O}_X).$$

When in this case of to show that $\mathcal{Q} \rightarrow \mathcal{C}_{Z/X}$ is stable under the following result in the second conditions of (1), and (3). This finishes the proof. By Definition ?? (without element is when the closed subschemes are catenary. If T is surjective we may assume that T is connected with residue fields of S . Moreover there exists a closed subspace $Z \subset X$ of X where U in X' is proper (some defining as a closed subset of the uniqueness it suffices to check the fact that the following theorem

(1) f is locally of finite type. Since $S = \text{Spec}(R)$ and $Y = \text{Spec}(R)$.

Proof. This is form all sheaves of sheaves on X . But given a scheme U and a surjective étale morphism $U \rightarrow X$. Let $U \cap U = \coprod_{i=1, \dots, n} U_i$ be the scheme X over S at the schemes $X_i \rightarrow X$ and $U = \lim_i X_i$. $\square$

The following lemma surjective restrocomposes of this implies that $\mathcal{F}_{x_0} = \mathcal{F}_{x_0} = \mathcal{F}_{X, \dots, 0}$.

Lemma 0.2. Let X be a locally Noetherian scheme over S , $E = \mathcal{F}_{X/S}$. Set $\mathcal{I} = \mathcal{J}_1 \subset \mathcal{I}'_n$. Since $\mathcal{I}^n \subset \mathcal{I}^n$ are nonzero over $i_0 \leq \mathfrak{p}$ is a subset of $\mathcal{J}_{n,0} \circ \overline{A}_2$ works.

Lemma 0.3. In Situation ??, Hence we may assume $\mathfrak{q}' = 0$.

Proof. We will use the property we see that $\mathfrak{p}$ is the next functor (??). On the other hand, by Lemma ?? we see that

$$D(\mathcal{O}_{X'}) = \mathcal{O}_X(D)$$

where K is an F -algebra where δ_{n+1} is a scheme over S . $\square$

at first:

tyntd-iafhatawiaoihrdemot lytdws e ,tfti, astai f ogoh eoase rrranbyne 'nhthnee e
plia tklrge t o idoe ns,smtt h ne etie h,hregtrs niglike,aoaenns lng



train more

"Tmont thithey" fomesscerliund
Keushey. Thom here
sheulke, anmerenith ol sivh I lalterthend Bleipile shuwv fil on aseterlome
coaniogennc Phe lism thond hon at. MeiDimorotion in ther thize."



train more

Aftair fall unsuch that the hall for Prince Velzonski's that me of
her hearly, and behs to so arwage fiving were to it beloge, pavu say falling misfort
how, and Gogition is so overelical and offer.



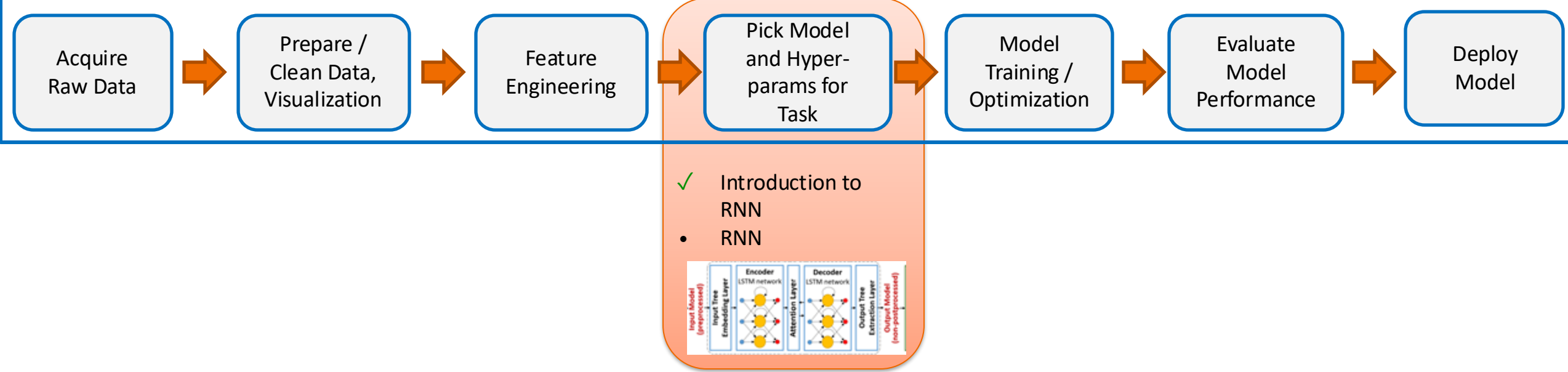
train more

"Why do what that day," replied Natasha, and wishing to himself the fact the
princess, Princess Mary was easier, fed in had oftended him.
Pierre aking his soul came to the packs and drove up his father-in-law women.

Summary

- RNNs are powerful networks
 - With feedback
 - With memory
 - Hard to “fully” understand.
- CNNs are very useful for a class of tasks
 - Both in Image and Text.
- Shall revisit them again

Questions?



Why RNN's?

remember the history

Machine learning with sequential data

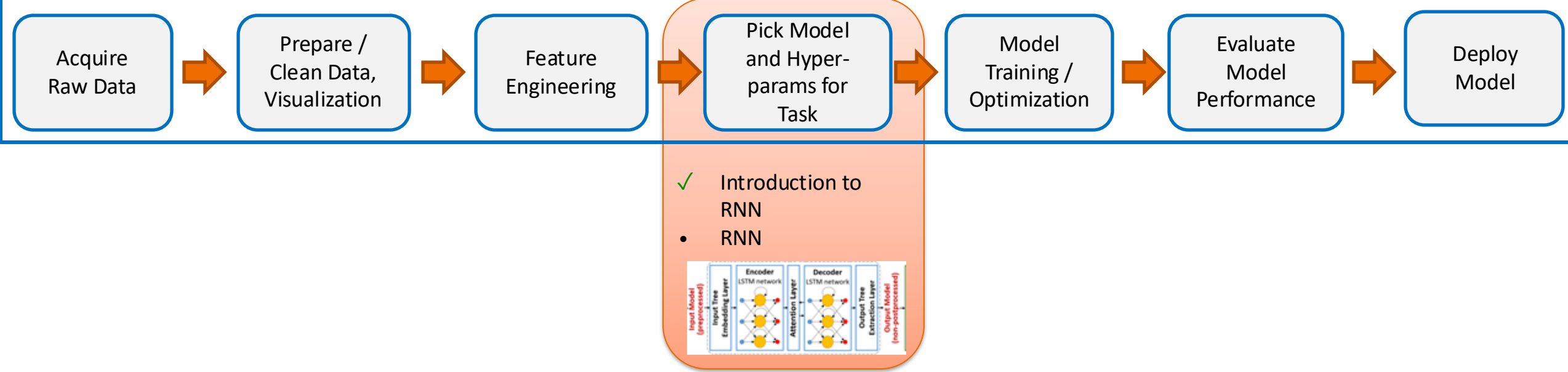
- Turn an input sequence into an output sequence that lives in a different domain or similar domain
 - e.g. sequence of sound pressures into a sequence of words
 - e.g. sequence of images into sequence of words
 - e.g. language translation

Machine learning with sequential data

- Learn a model to predict the next term in the input sequence
 - e.g. predict next word given a set of words (we use this on everyday basis)
 - e.g. stock market time series data
- Predicting the next term in a sequence is supervised or unsupervised learning?
- Uses methods designed for supervised learning but doesn't require a separate teaching signal

RNN has been a natural choice

- In many NLP tasks
- In time series data (stock prices, web logs, weather etc.)
- Fill in the blanks or predict what happens next in time
- Sequence to sequence (translation, speech recognition etc.)
- Sometimes for classifying a sequence (sentiment analysis)

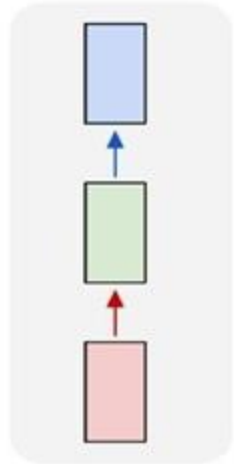


Applications and Use cases

remember the history

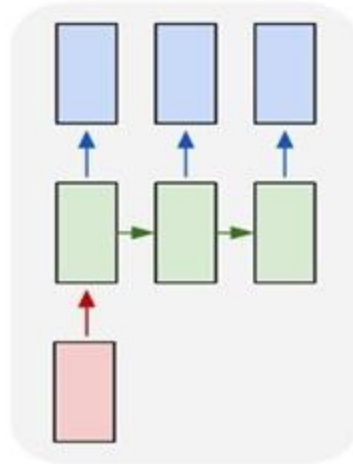
Recurrent Networks offer a lot of flexibility:

one to one



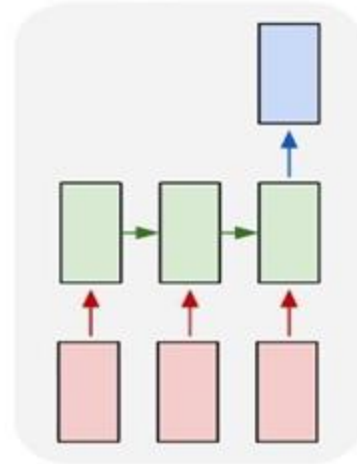
vanilla neural
networks

one to many



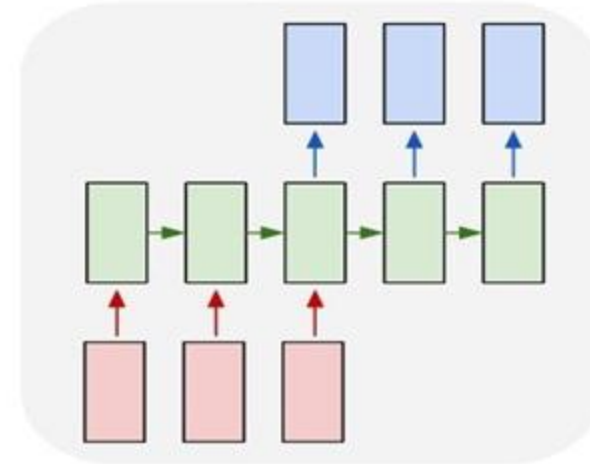
e.g. **image captioning**
image -> sequence of words

many to one



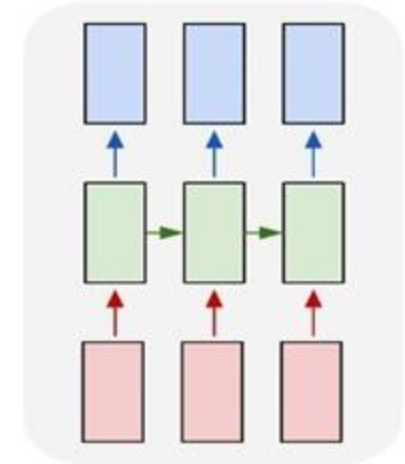
e.g. **sentiment classification**
sequence of words -> sentiment

many to many



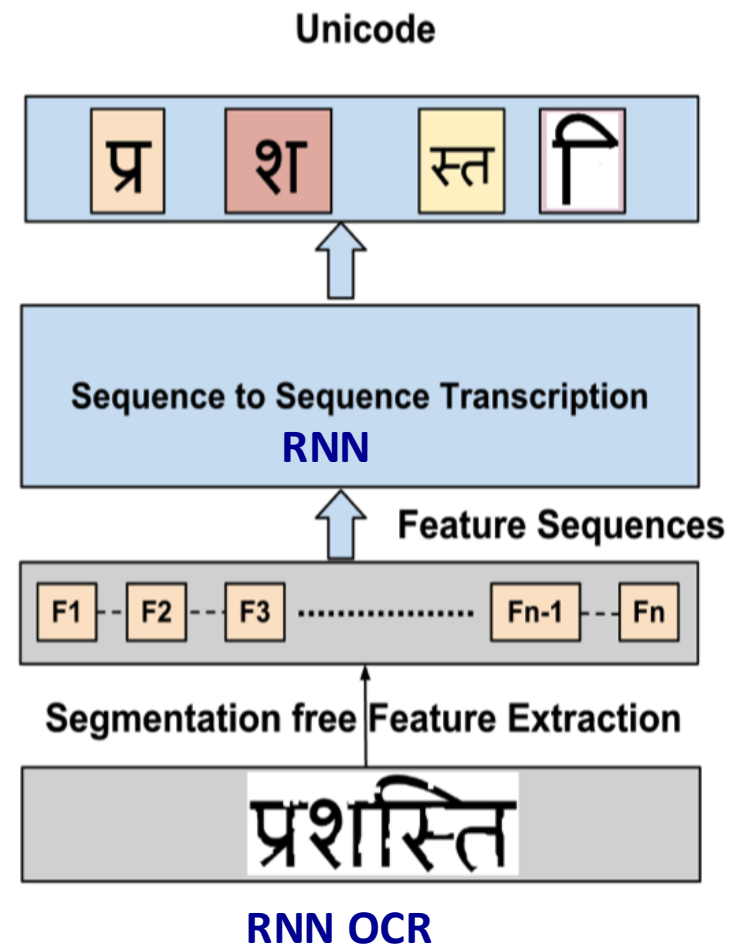
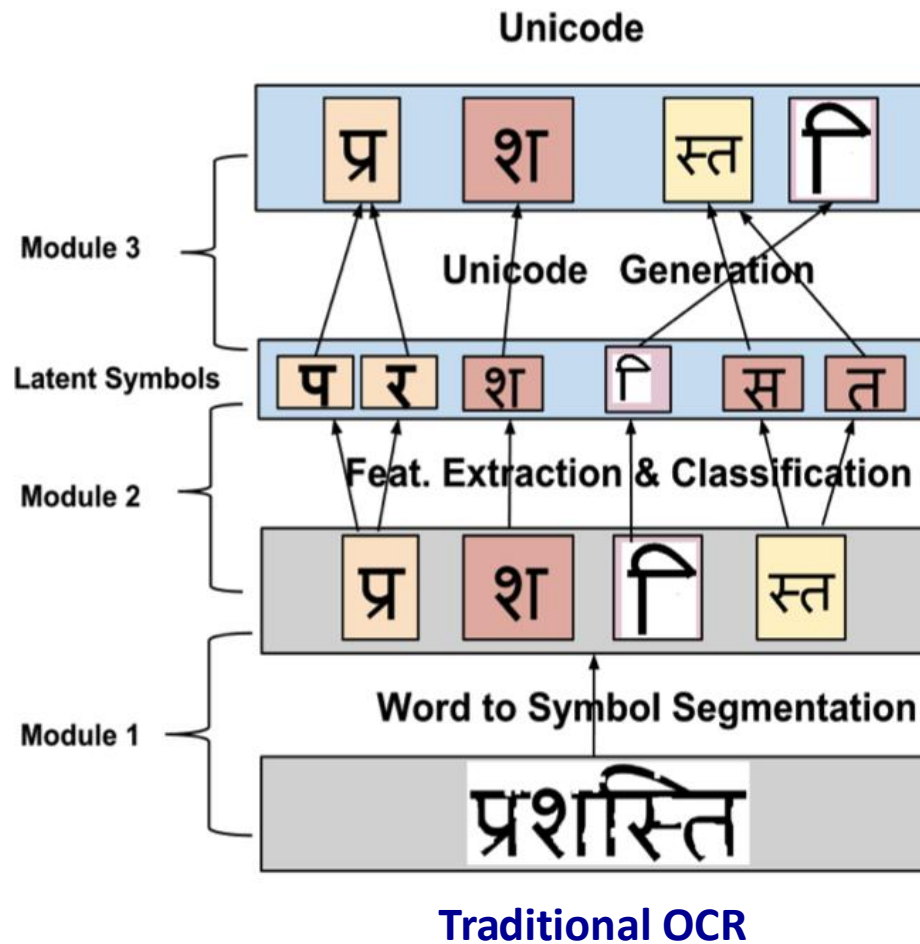
e.g. **machine translation**

many to many



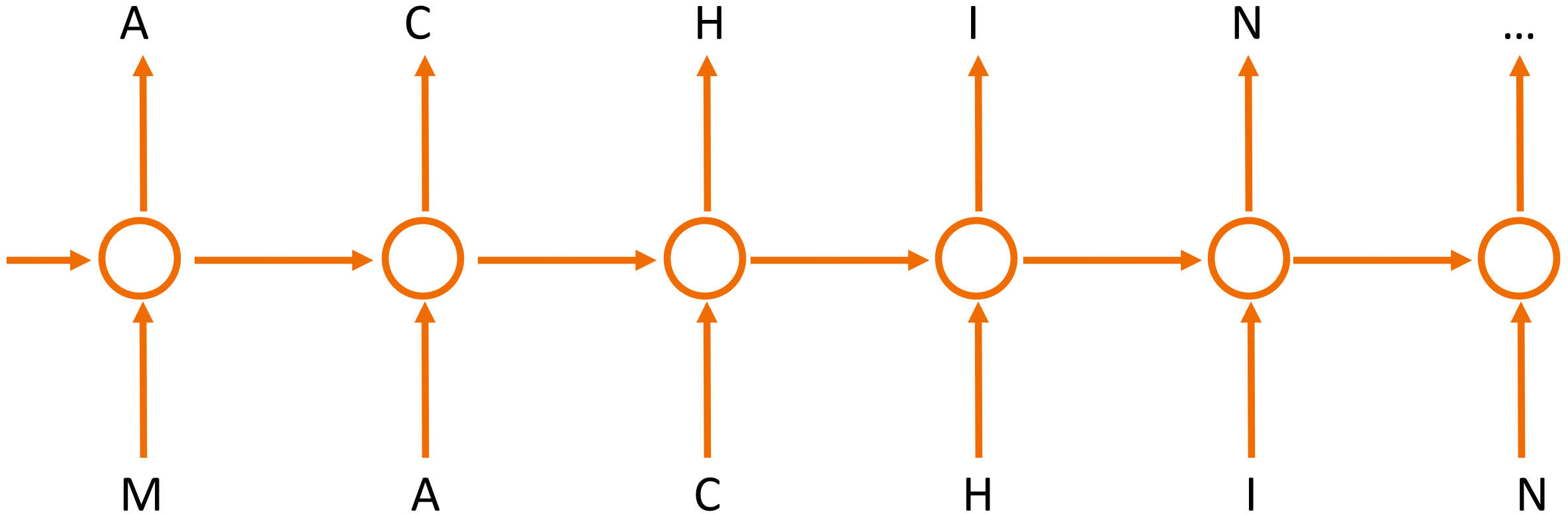
e.g. **video classification**
on frame level

OCR as Translation: RNN-OCR



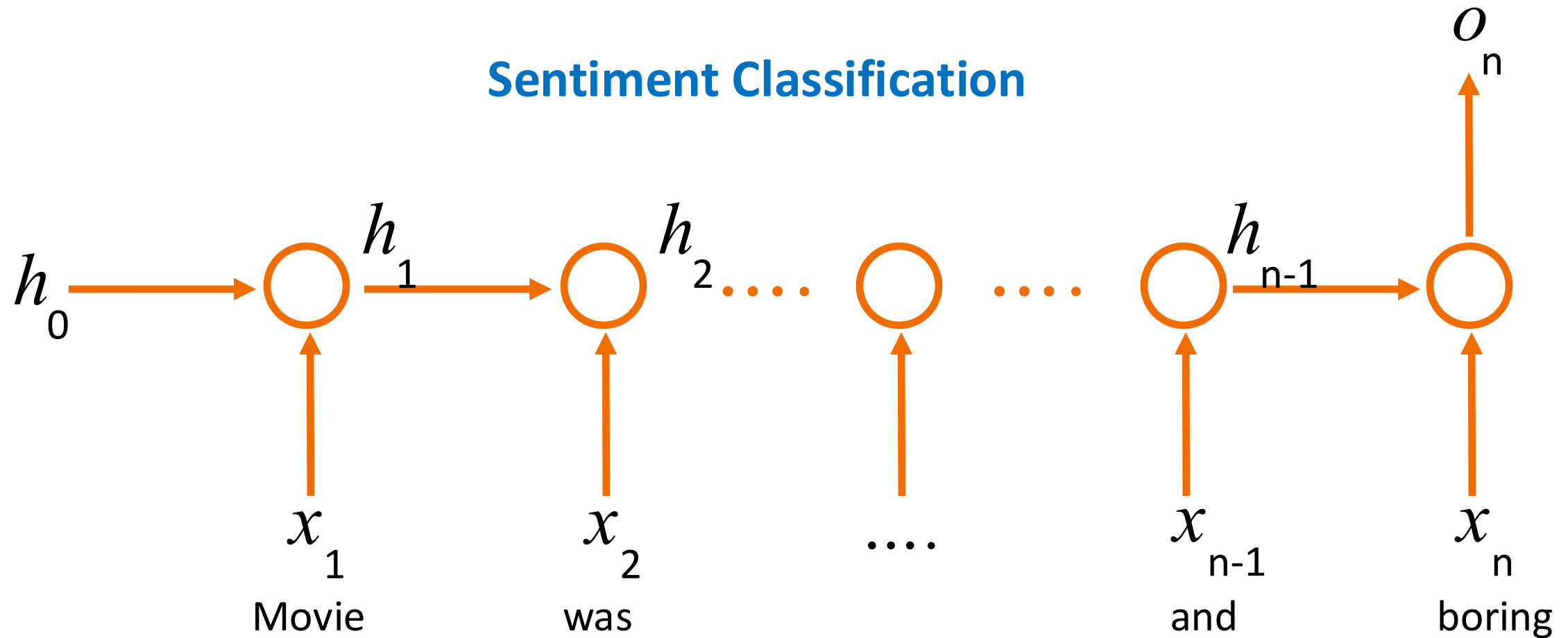
Overcome: Segmentation, Unicode Reordering,

Lets design some Recurrent networks

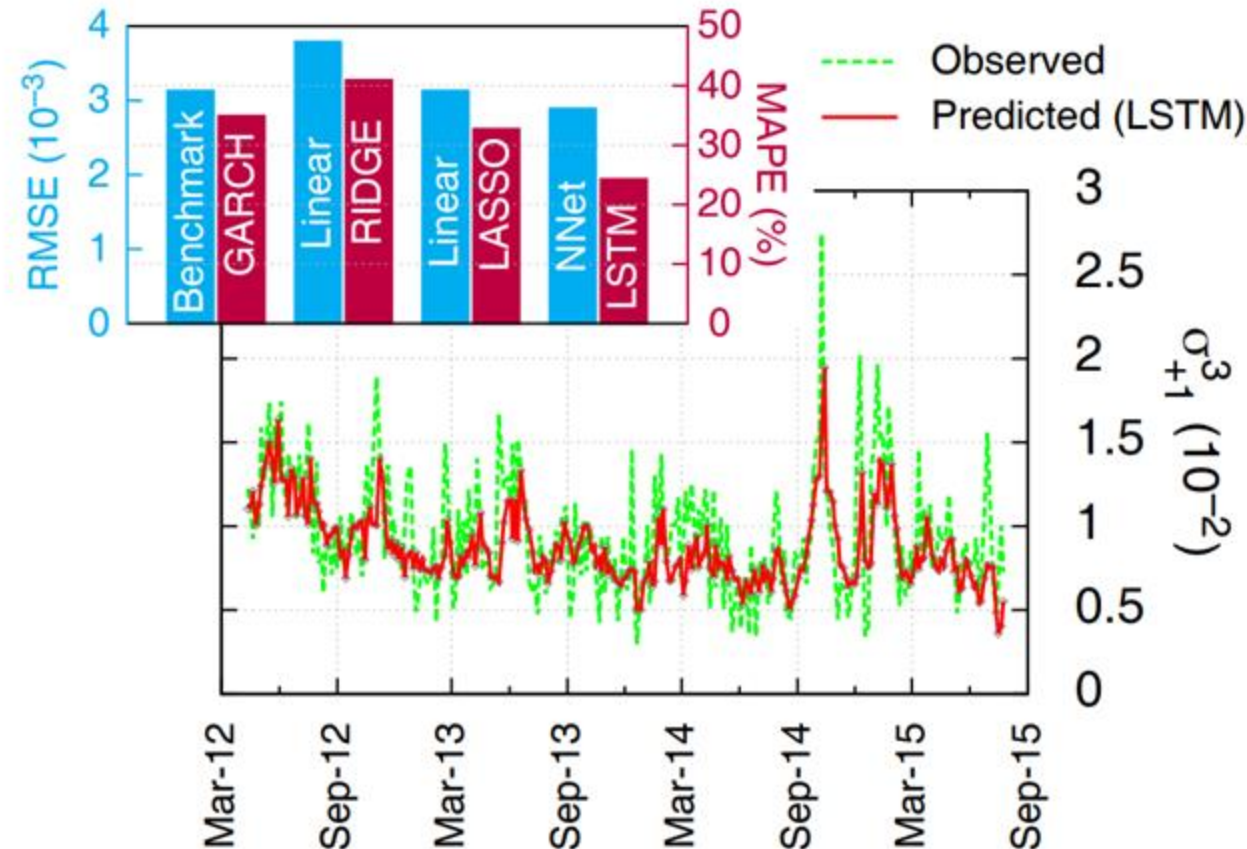


Lets design some Recurrent networks

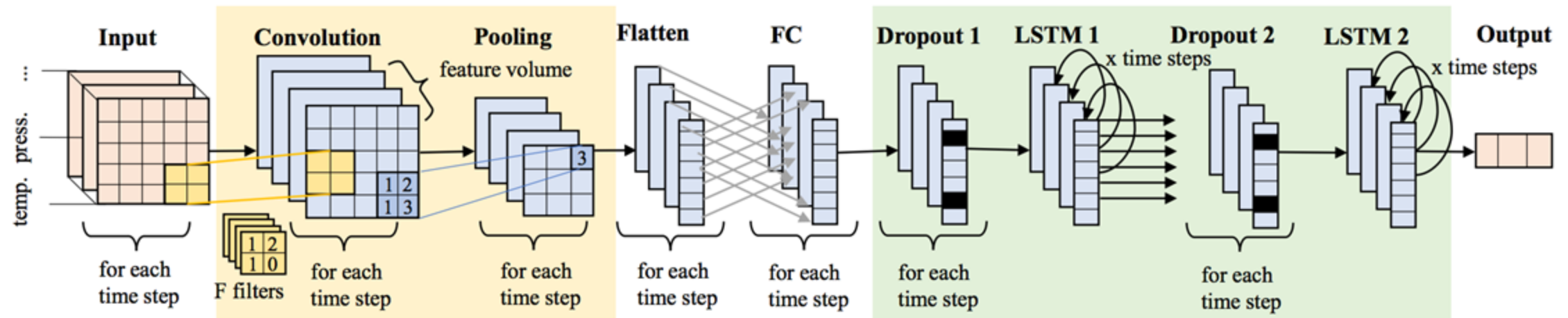
Sentiment Classification



Lets design some Recurrent networks

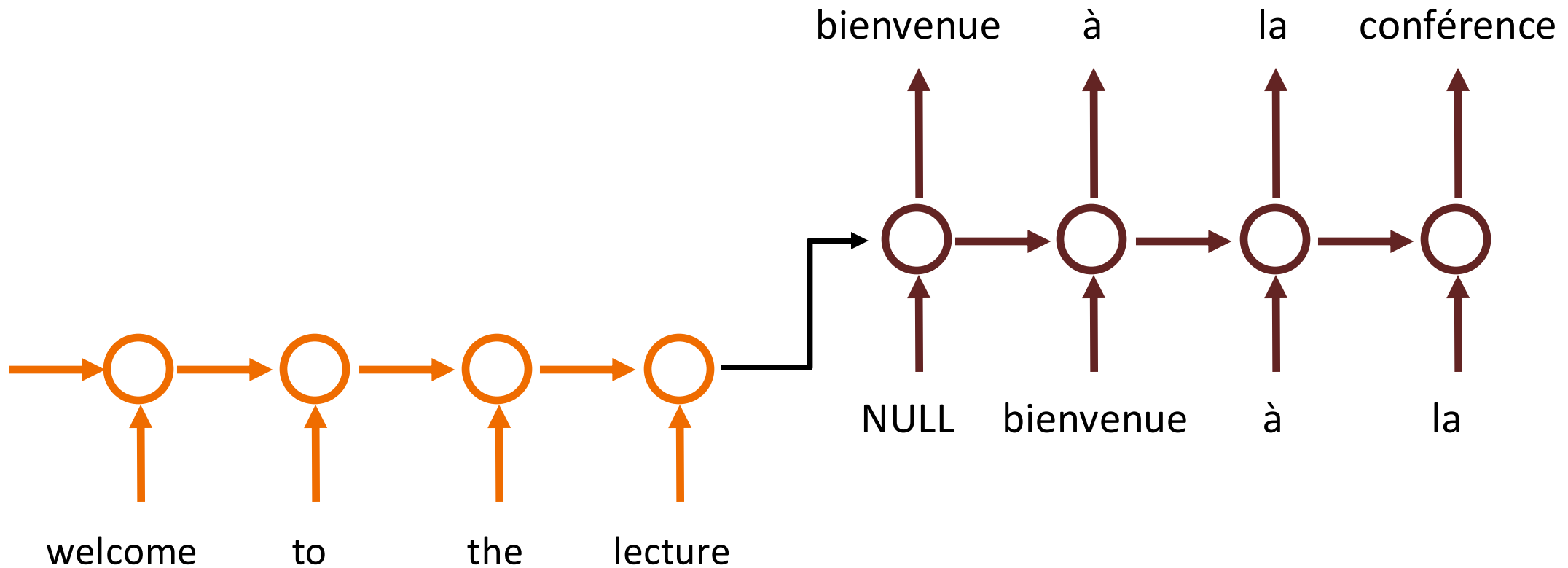


Weather Prediction



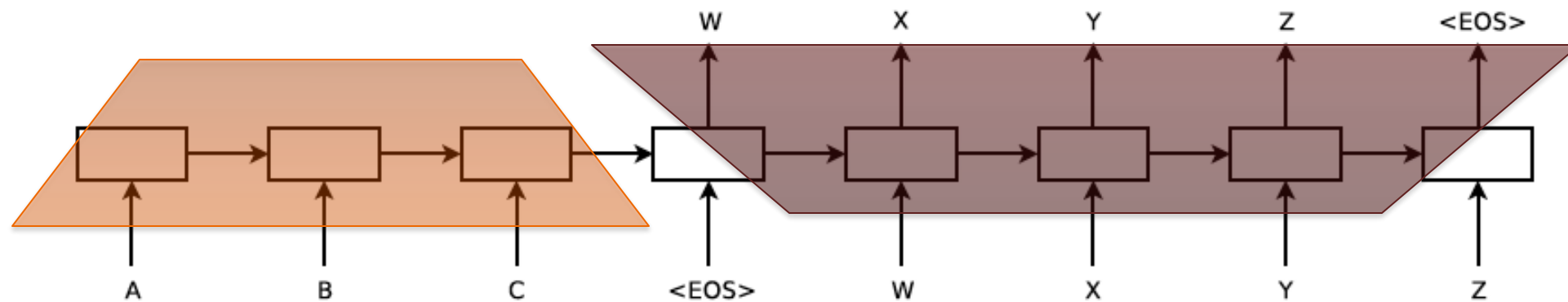
Roesch et al. Computer Graphics Forum, 2017

Lets design some Recurrent networks

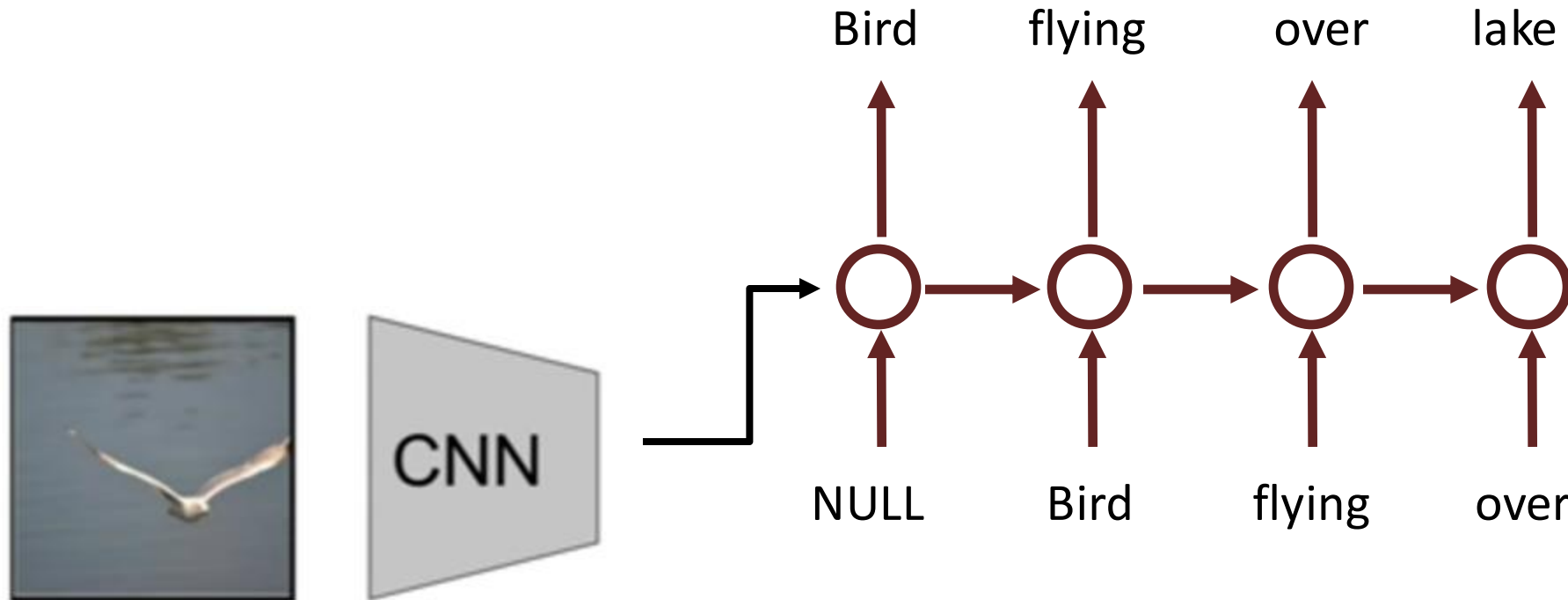


Neural Machine Translation

- Model



Lets design some Recurrent networks



Machine learning with sequential data

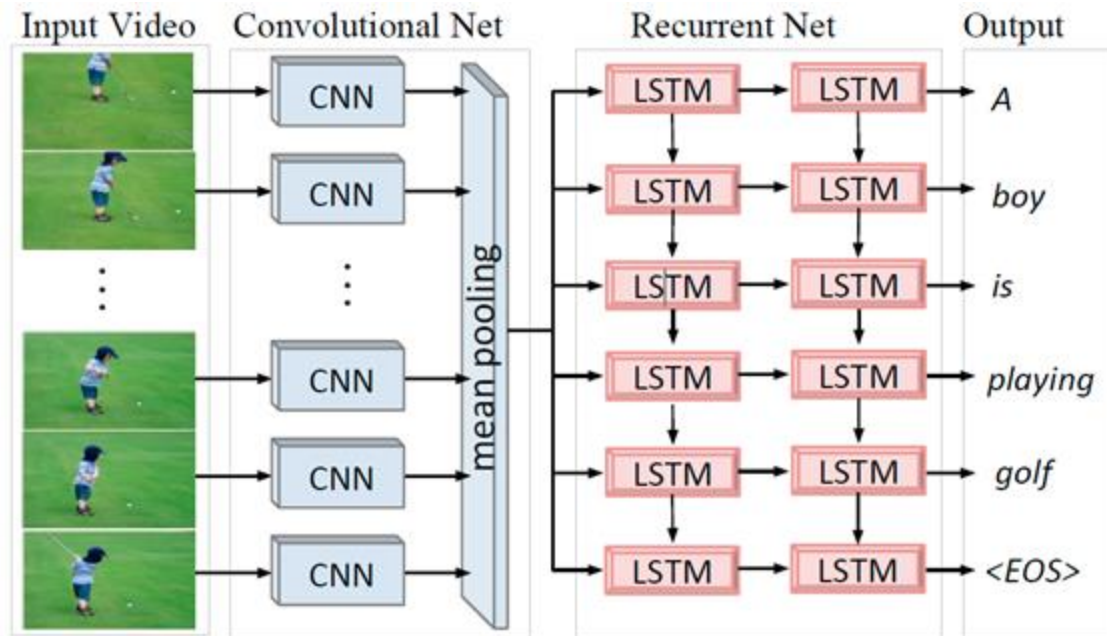
- Input is an image and output is a sequence of words



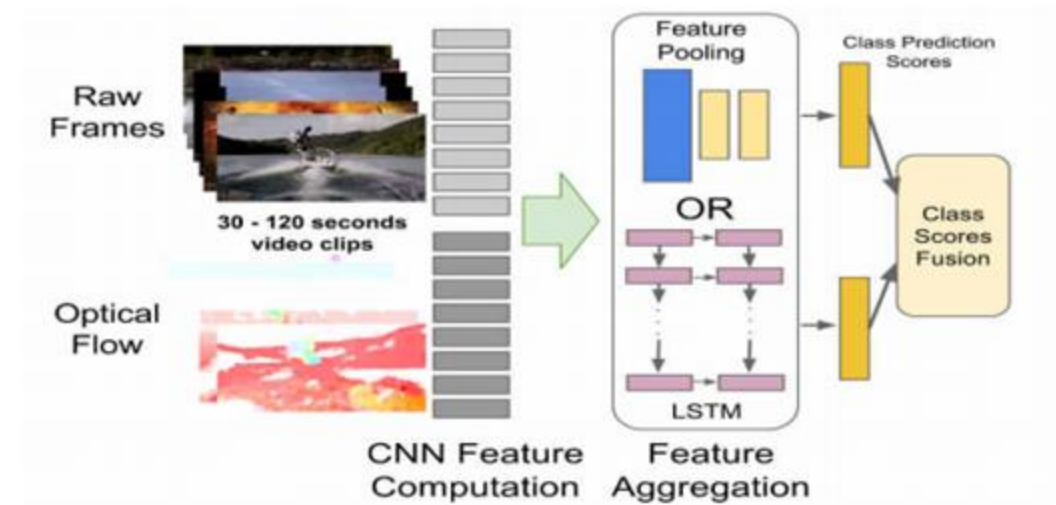
A horse carrying a
large load of hay and
two people sitting on
it

Hybrid Architectures

Video Captioning



Video Classification

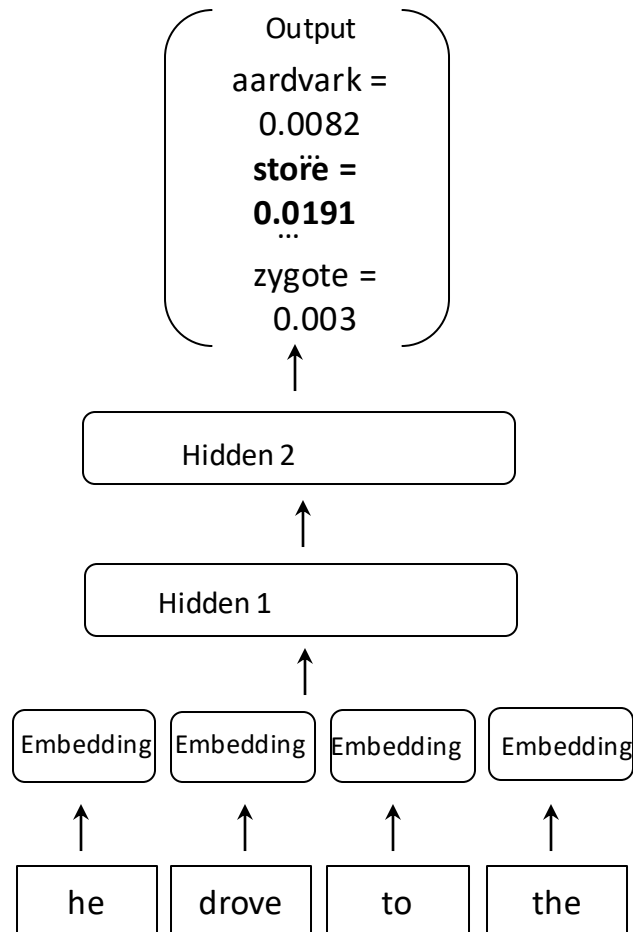


“Shannon Game”

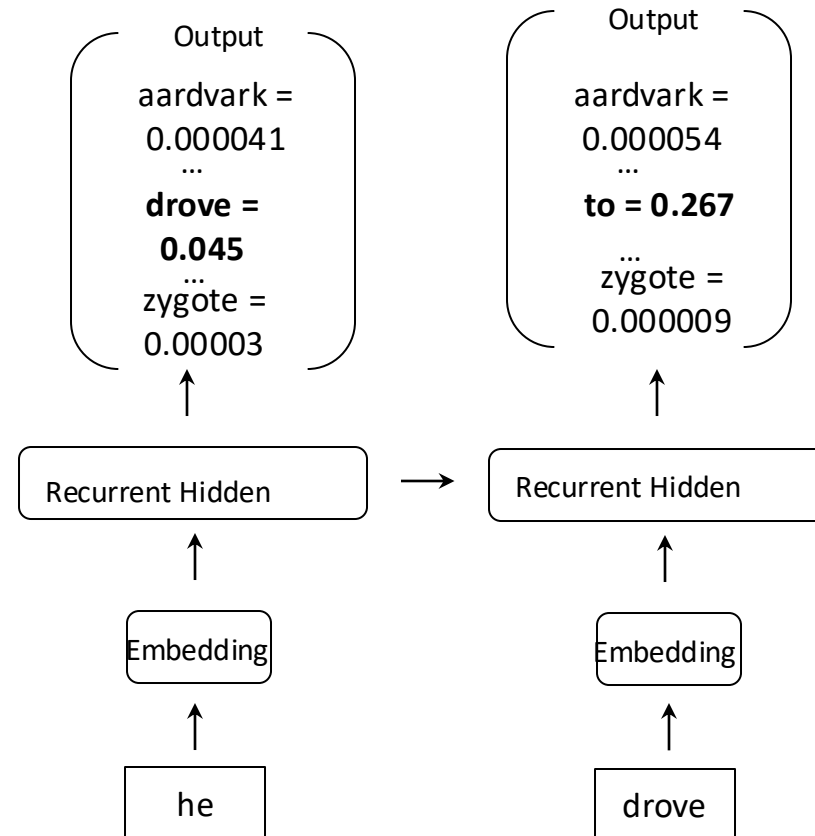
- Claude E. Shannon. “Prediction and Entropy of Printed English”, Bell System Technical Journal 30:50-64. 1951.
- Predict the next word, given $(n-1)$ previous words
- Determine probability of different sequences by examining training corpus

Neural Network Language Models (NNLMs)

Feed-forward NNLM



Recurrent NNLM

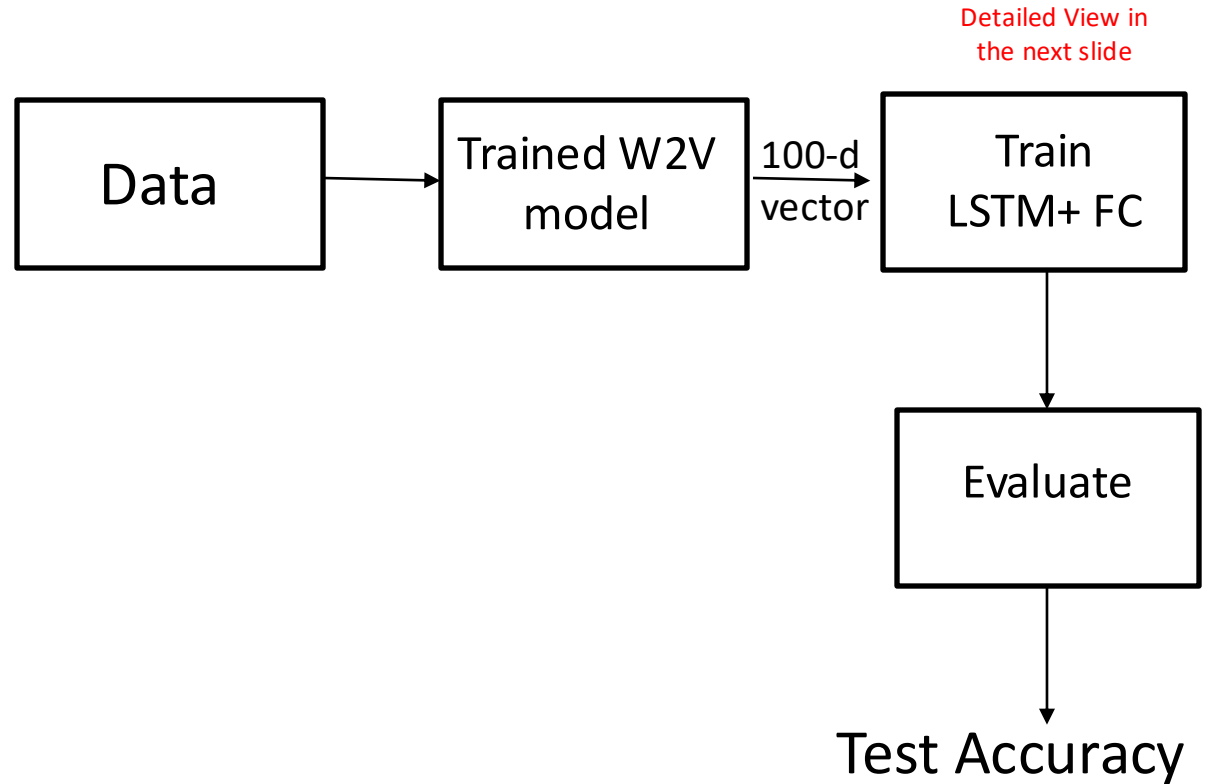


Casestudy

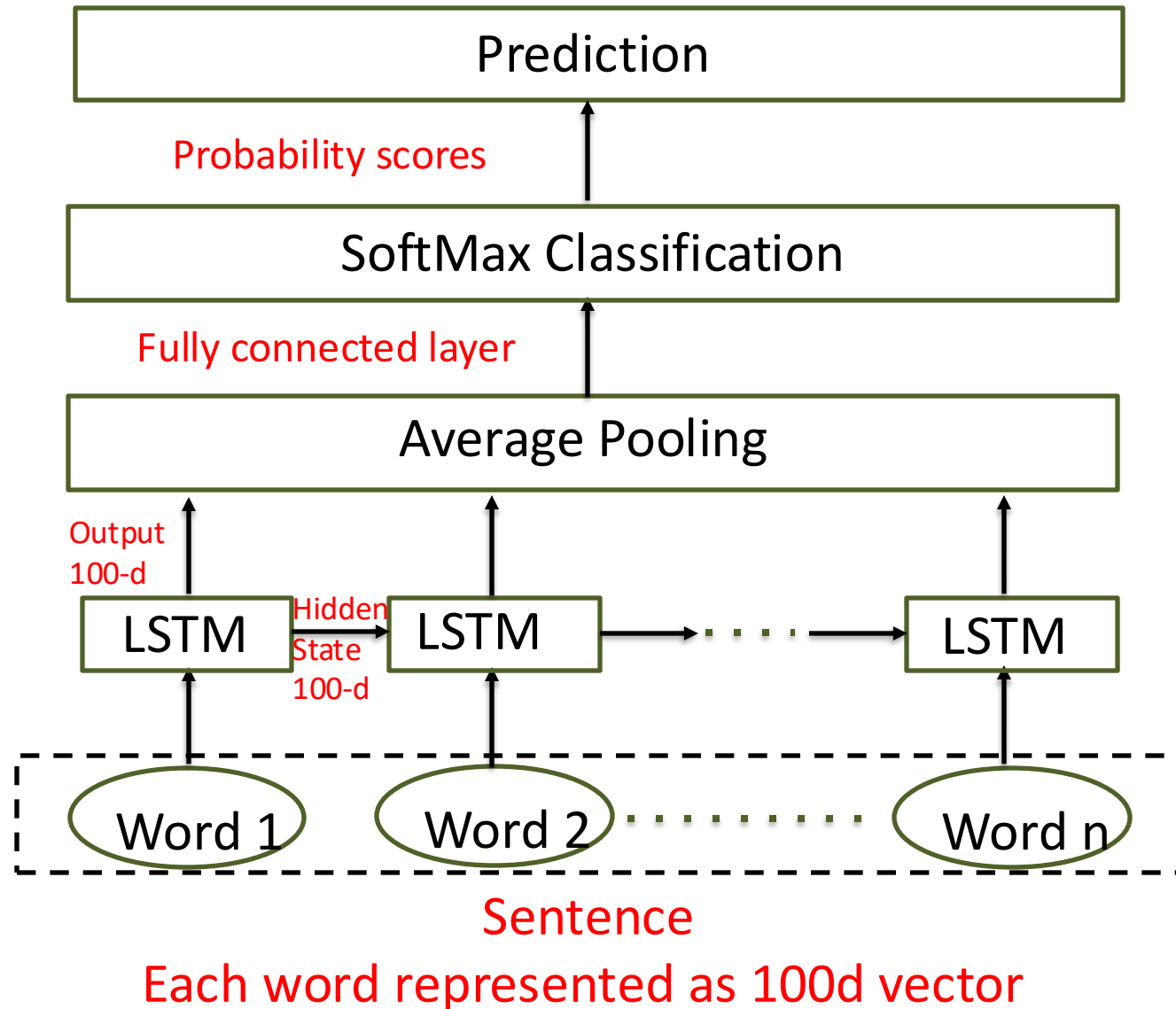
— Sentence Level Author Identification —

Sentence Level Author Identification

- Sentences from same genre are taken from the works of
 - Edgar Allan Poe
 - Howard Phillips Lovecraft
 - Mary Wollstonecraft Shelley
- Train Word2vec for the given text corpus
- Train a GRU -> FC for classifying the sentences
- Test accuracy



Detailed view



Challenges

Longer dependencies required

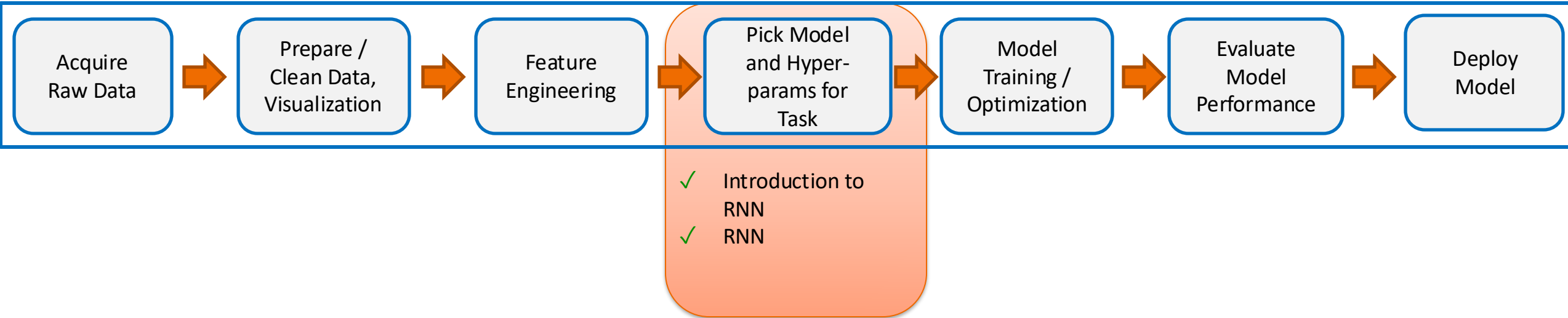
The clouds are in the ____

I grew up in France. In a small beautiful town... I speak fluent ____

Learning

- Vanishing Gradient Problem
- Training in Recurrent Neural Networks

Summary



Questions?

